FIITJEE

ALL INDIA TEST SERIES

CONCEPT RECAPITULATION TEST – II

JEE (Advanced)-2019

PAPER – 2

ANSWERS, HINTS & SOLUTIONS

Physics

PART – I

SECTION – A

1. ABCD

Sol. Charge on capacitor at time t is

$q = Q_0 \cos \omega t$

Where $\omega = \frac{1}{\sqrt{LC}}$

At $\omega t = \frac{\pi}{2}, q = 0$

$\Rightarrow \ t = \frac{\pi}{2}\sqrt{LC}$

By conservation of energy

$\frac{Q_0^2}{2C} = \frac{1}{2}Li_{max}^2$

$\Rightarrow \ i_{max} = \frac{Q_0}{\sqrt{LC}}$

2. ACD

Sol. Apply Gauss law and property of conductor.

3. AD

Sol. $V = \frac{-GM}{2R^3}[3R^2 - r^2]$, inside the earth.

4. ABCD

Sol. Let elongation in spring A, B and C be x_1, x_2 and x_3 respectively.
Considering spring forces and constraint relations

$x_2 = 4x_3$ …(i)

$x_2 = 2x_1$ …(ii)

and $x_1 + 2x_2 + x_3 = x$ …(iii)

$\Rightarrow \; x_1 = \left(\frac{2}{11}\right)x \;\; ; \;\; x_2 = \left(\frac{4}{11}\right)x \;\; ; \;\; x_3 = \left(\frac{1}{11}\right)x$

Also, $F = 2K\left(\frac{x}{11}\right)$

$\Rightarrow \; T = 2\pi\sqrt{\frac{11m}{2k}}$

5. AB

Sol. For microwaves,

$\lambda = \frac{c}{f} = \frac{3\times10^8}{10^6} = 300$ m.

$\Delta x = d \sin\theta$

$\phi = \frac{2\pi}{\lambda} d\sin\theta = \frac{2\pi}{300}(150\sin\theta) = \pi \sin\theta$

$I_\theta = I_0 \cos^2\left(\frac{\pi\sin\theta}{2}\right)$

6. ABC

Sol. $T = \frac{2\sqrt{2}}{10} = \frac{\sqrt{2}}{5}$; $R = \sqrt{(1)^2 + (1)^2}$ $T = \frac{2}{5}$

$H = \frac{\left(\sqrt{2}\right)^2}{2g} = \frac{1}{10}$

7. C

Sol. Optical path difference at any point P on the screen, $\delta(P) = S_2P - S_1P - (\mu - 1)t$ and the intensity on the screen, at point $P = 4I_o \times \cos^2 \frac{\pi}{\lambda} \times \delta P$.

8. A

Sol. (A) $A \to B$: V ↓, P constant → T↓, U↓ and ΔW is –ve, $\Delta Q < 0$, $\Delta U < 0$
(B) $B \to C$: V is same, P ↓ T ↓, U↓, $\Delta Q < 0$, $\Delta U < 0$
No work is done.
(C) $C \to D$: V ↑ ⇒ T↑, $\Delta U > 0$, $\Delta Q > 0$, $\Delta W > 0$
(D) $D \to A$: V decrease so $\Delta W < 0$
$\therefore \; T_D = T_A$ $\therefore \; \Delta U = 0$
and then $\Delta Q < 0$

9. B

Sol. Mean value $= \dfrac{12.5+12.3+11.8+12.4+12.2+12.6}{6} = 12.3$

Mean absolute error = $\dfrac{|\Delta x_1| + ... + |\Delta x_6|}{6}$

Relative error = $\dfrac{\text{Mean absolute error}}{\text{Mean value}}$

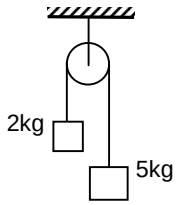
10. D

Sol. According to Lenz's law induced current flows in such a way that it is opposing the change in magnetic flux.

SECTION – D

11. 00028.00

Sol. $T = \dfrac{2m_1m_2g}{m_1+m_2} = \dfrac{2\times2\times5\times9.8}{7} = 28$ N



12. 00000.10

Sol. Given, $\dfrac{1}{2}mv^2 = \dfrac{1}{2}mu^2(1-0.19) = \dfrac{1}{2}mu^2(0.81)$

$\therefore$ v = 0.9 u

$\therefore$ $\dfrac{\Delta v}{u} = 0.10$

13. 00000.09

Sol. $\varepsilon = vB\ell = 3\times10^6\times10\times3\times10^{-9} = 9\times10^{-2} \approx 0.09$ V

14. 00000.23

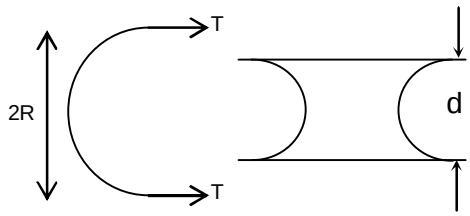
Sol. $\mu = \dfrac{\text{velocity in air}}{\text{velocity in medium}}$

15. 00001.00

Sol. The shape of water layer between the two plates is shown in the figure.

Thickness *d* of the film = 0.12 mm = 0.012 cm.

Radius *R* of cylindrical face $= \dfrac{d}{2}$.



$F = T(2l) = P(l\times2R)$

$P = \dfrac{T}{R}$

Pressure difference across the cylindrical surface $= \dfrac{T}{R} = \dfrac{2T}{d}$.

Area of each plate wetted by water = *A*.
Force *F* required to separate the two plates is given by

F = pressure difference × area $= \frac{2T}{d}A = \frac{2\times 75\times 8}{0.012} = 1N$

16. 00008.00

Sol. $\frac{4\ell_1}{3} = \frac{2\ell_2}{4}$

17. 00000.00

Sol. $dw = \vec{E}.d\vec{r} = 0$

18. 00004.00

Sol. $X_{cm} = \frac{\sigma\pi R^2(O) + \sigma\pi\left(\frac{R}{2}\right)^2\frac{R}{2} + (-\sigma)\pi\left(\frac{R}{2}\right)^2\left(-\frac{R}{2}\right)}{\sigma\pi R^2 + \sigma\pi\left(\frac{R}{2}\right)^2 + (-\sigma)\pi\left(\frac{R}{2}\right)^2}$

Chemistry

PART – II

SECTION – A

19. ABD

Sol. O_2^+ contains unpaired electron in π-antibonding M.O and N_2^+ in σ_{2p} orbital N_2^{2-} contains unpaired electron in π-antibonding M.OS.

20. ABC

Sol. Rate = $k[X][Y]^0 \Rightarrow 2 \times 10^{-8} = k[0.1]\,[0.1]^0 \Rightarrow k = 2 \times 10^{-7} s^{-1}$.

21. ABD

Sol. When Cl is directly attached to benzene, +R effect of chlorine is observed. Which increases the bond order of C – Cl bond.

22. BCD

Sol. Elimination takes place in option(A).

23. AC

Sol. Azeotropic mixture has fixed composition.

24. ACD

Sol. $a = 2\left(r_{Na^+} + r_{Cl^-}\right) = 2(95 + 181) = 552\,pm$

25. B

Sol. $2LiOH \xrightarrow{\Delta} Li_2O + H_2O$

26. C

Sol. In E_2 mechanism, loss of Hydrogen takes place from that carbon atoms which are adjacent to the carbon atoms that hold chlorine. Carbocations are formed in E_1 mechanism.

27. D

Sol.

$$CH_3CONH_2 \begin{cases} \xrightarrow{H_2O/H^+} CH_3COOH + NH_4^+ \\ \xrightarrow{H_2O/OH^-} CH_3COO^- + NH_3 \end{cases} \qquad (CH_3CO)_2O \xrightarrow{H_2O} 2CH_3COOH$$

28. A

Sol. Fact based

SECTION – D

29. 00400.00

Sol. $K_{P_1} = p_{SO_2} \times p_{SO_3} = 900$

$\therefore p_{SO_2} = p_{SO_3} = 30\,\text{cm of Hg}$

$$K_{P_2} = \frac{p_{SO_3}^2}{p_{SO_2}^2 \times p_{O_2}} = \frac{30}{30 \times p_{O_2}} = 25 \times 10^{-4}$$

$$\therefore p_{O_2} = \frac{1}{25 \times 10^{-4}} = 400 \text{ cm of Hg}$$

30. 00208.30

Sol. $K_{sp} = 4s^3 = 4 \times 10^{-3}$

$\therefore s = 10^{-1} = 0.1 \text{ mol } L^{-1}$

In 10 L $\rightarrow$ 1 mole dissolves

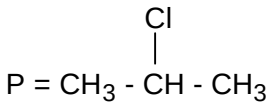
Mass of 1 mole = 208.3 g

31. 00138.00

Sol. P = K_2CO_3, Q = O_2, R = KOH, S = H_2O_2. Heaviest product is K_2CO_3.

32. 00157.00

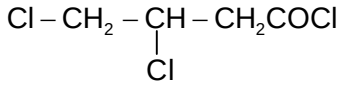
Sol.



P = $CH_3 - CH - CH_3$

33. 00175.50

Sol. The product is



34. 00226.50

Sol. The complex is $[Cr(NH_3)_4Cl_2]Cl$.

35. 95735.71

Sol. $W = -2.303nRT \log \frac{V_2}{V_1} = -2.303 \times 1 \times 8.314 \times 5000 \times \log \frac{1/10}{1} = 95735.71$

36. 00390.55

Sol. $$\Lambda_m^0 CH_3COOH = \frac{\left(\Lambda_m^0 Ca(CH_3COO)_2 + 2\Lambda_m^0 HCl - \Lambda_m^0 CaCl_2\right)}{2}$$

$$= \frac{200.8 + 2(425.95) - 271.6}{2} = 390.55$$

Mathematics

PART – III

SECTION – A

37. ACD

Sol. $\tan\frac{\pi}{7}+2\tan\frac{2\pi}{7}+4\tan\frac{4\pi}{7}+8\cot\frac{8\pi}{7}=\cot\frac{\pi}{7}\tan$

$\tan\theta+2\tan2\theta+4\tan4\theta+8\cot8\theta=\cot\theta$ [when $\theta=\frac{\pi}{7}$]

(a) $\operatorname{cosec}2\theta+\cot2\theta=\frac{1+\cos2\theta}{\sin2\theta}=\cot\theta=\cot\frac{\pi}{7}$ (where $\theta=\frac{\pi}{7}$)

(b) $\tan\frac{\pi}{14}-\cot\frac{\pi}{14}=-2\cot\frac{\pi}{7}$

(c) $\frac{\sin\frac{2\pi}{7}}{1-\cos\frac{2\pi}{7}}=\frac{2\sin\frac{\pi}{7}\cos\frac{\pi}{7}}{2\sin^2\frac{\pi}{7}}=\cot\frac{\pi}{7}$

(d) $\frac{\left(1+\cos\frac{2\pi}{7}\right)+\cos\frac{\pi}{7}}{2\sin\frac{\pi}{7}\cos\frac{\pi}{7}+\sin\frac{\pi}{7}}=\frac{2\cos^2\frac{\pi}{7}+\cos\frac{\pi}{7}}{2\sin\frac{\pi}{7}\left(\cos\frac{\pi}{7}+1\right)}=\cot\frac{\pi}{7}$

38. ABCD

Sol. (a) We have $f(x)=\frac{\sec x}{\cos x}-\frac{\tan x}{\cot x}$, $g(x)=\frac{\cos x}{\sec x}+\frac{\sin x}{\operatorname{cosec} x}$

Clearly both $f(x)$ and $g(x)$ are identical functions as $x\neq\frac{k\pi}{2}\forall k\in I$.

(b) As $x^2-4x+5=(x-2)^2+1>0$

Hence $f(x)=1\forall x\in R$.

Also $\cos^2 x+\sin^2\left(x+\frac{\pi}{2}\right)>0$

Hence $g(x)=1\forall x\in R$ g.

$\Rightarrow f(x)$ and $g(x)$ are identical.

(c) $f(x)=e^{\ln\left(x^2+3x+3\right)}$

As $x^2+3x+3=\left(x+\frac{3}{2}\right)^2+\frac{3}{4}>0\forall x\in R$.

Hence $f(x)=x^2+3x+3\forall x\in R$

$\Rightarrow f(x)$k is identical to $g(x)$

(d) We have $f(x)=\frac{\sin x}{\sec x}+\frac{\cos x}{\operatorname{cosec} x}$, $g(x)=\frac{2\cos^2 x}{\cot x}$

Clearly both $f(x)$ and $g(x)$ are identical functions as $x\neq\frac{k\pi}{2}\forall k\in I$.

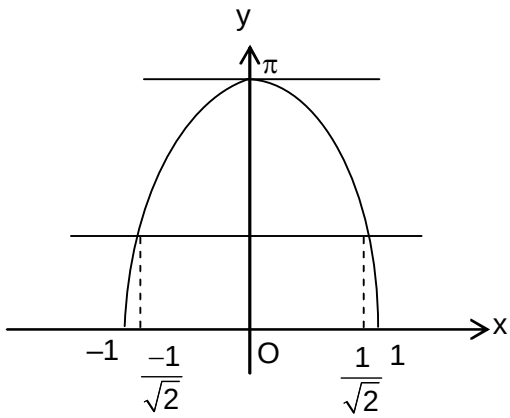
39. AC

Sol. Differential coefficient of $f(x)$ w.r.t.

$\sin^{-1}x = -2\dfrac{x}{|x|}$ at $x = \dfrac{1}{2}$ it is –2

$f(0^+) = -2; f'(0^-) = 2$

$\Rightarrow$ f is not derivable at x = 0



40. ABC

Sol. $I_1 = \int_0^{\pi/2} \cos(\pi \sin^2 x)\,dx$

$\int_a^b f(x)\,dx = \int_a^b f(a+b-x)\,dx$

$I_1 = \int_0^{\pi/2} \cos(\pi \cos^2 x)\,dx$

On adding

$2I_1 = \int_0^{\pi/2} \cos(\pi \sin^2 x) + \cos(\pi \cos^2 x)\,dx$

$= \int_0^{\pi/2} 2\cos\left(\frac{\pi}{2}\right).\cos\left(\frac{\pi}{2}\cos 2x\right)dx = 0$

$I_1 = 0$ ……..(i)

$I_2 = \int_0^{\pi/2} \cos\left(\pi(1-\cos 2x)\,dx = -\int_0^{\pi/2} \cos(\pi\cos 2x)\,dx\right)$

$= -\frac{1}{2}\int_0^{\pi} \cos(\pi\cos t)\,dt$ [Put $2x = t$]

$= -\frac{2}{2}\int_0^{\pi/2} \cos(\pi\cos t)\,dt = -I_3$

$\therefore I_2 + I_3 = 0$ …….(ii)

Hence, $I_1 + I_2 + I_3 = 0$

41. BCD

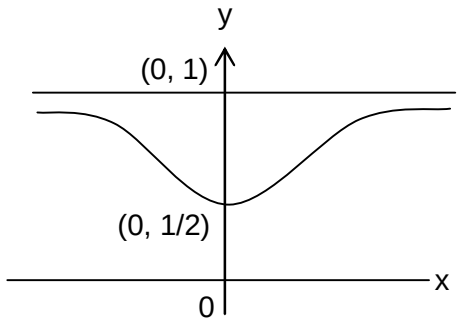
Sol. Domain is $x \in R$

Also $f(x) = \left[\cos\left(\tan^{-1}(\sin\theta)\right)\right]^2$

where $\cot\theta = x$

$= \left[\cos\left(\tan^{-1}\left(\frac{1}{\sqrt{1+x^2}}\right)\right)\right]^2$

$= (\cos\phi)^2$ where $\tan\phi = \dfrac{1}{\sqrt{1+x^2}} = \left(\dfrac{\sqrt{1+x^2}}{\sqrt{2+x^2}}\right)^2$



$g(x) = \dfrac{1+x^2}{2+x^2} = 1 - \dfrac{1}{2+x^2}$

Range is $\left[\dfrac{1}{2}, 1\right)$; $f'(x) = \dfrac{2x}{\left(2+x^2\right)^2}$

Hence $f'(0) = 0$

Also $\lim\limits_{x\to\infty} f(x) = 1$

Hence (B), (C), (D)

42. ABC

Sol. $f(x) = \dfrac{\sin x}{x}$

43. A

Sol. (P) Let $S = \cos\theta + \cos 2\theta + \cos 4\theta + ... + \cos 10\theta$

$2\sin\theta . S = 2\sin\theta\left[\cos\theta + \cos 2\theta + ... + \cos 10\theta\right]$

$S = \dfrac{2\sin 6\theta\cos 5\theta}{2\sin\theta} = \dfrac{\sin n\theta\cos m\theta}{\sin\theta}$

$\Rightarrow n = 6$ and $m = 5$

(Q) $E = 9x\sin x + \dfrac{4}{x\sin x}$

[Note that $x\sin x > 0$ in $(0, \pi)$]

$E = \left(3\sqrt{x\sin x} - \dfrac{2}{\sqrt{x\sin x}}\right)^2 + 12$

$\therefore E_{min} = 12$ which occurs when $3x\sin x = 2$

$\Rightarrow x\sin x = \dfrac{2}{3}$

[Note that $x\sin x$ is continuous at x = 0 and attains the value $\dfrac{\pi}{2}$ which is greater than $\dfrac{2}{3}$ at $x = \dfrac{\pi}{2}$, hence it must take the $\dfrac{2}{3}$ in $(0, \pi)$]

(R) $f(x) = 2\left(\left(\sin x - 2\right)^2 + \dfrac{1}{2}\right)$

$\therefore M = 19$ and $m = 3$

Hence $\dfrac{M+8}{m} = \dfrac{19+8}{3} = 9$

(S) $\tan 9\theta = \frac{3}{4}$

$\sin 9\theta = \frac{3}{5}; \cos 9\theta = \frac{4}{5}$

Now given $\frac{3}{\sin 3\theta} - \frac{4}{\cos 3\theta}$

$= 2\frac{(3\cos 3\theta - 4\sin 3\theta)}{2\sin 3\theta \cos 3\theta}$

$= \frac{10\left[\frac{3}{5}\cos\theta - \frac{4}{5}\sin 3\theta\right]}{\sin 6\theta}$

$= 10\left[\frac{\sin 9\theta 3\theta - \cos 9\theta \sin 3\theta}{\sin 6\theta}\right] = 10\frac{\sin(9\theta - 3\theta)}{\sin 6\theta} = 10$

44. B

Sol. (P) $\lim_{h\to 0}\frac{f(3\cos h + 4\sin h - 2) - f(1)}{f(3e^h - 5\sec h + 4) - f(2)}\left(\frac{0}{0}\text{form}\right)$

$= \lim_{h\to 0}\frac{(-3\sin h + 4\cos h)f'(3\cos h + 4\sin h - 2)}{(3e^h - 5\sec h\tan h)f'(3e^h - 5\sec h + 4)} = \frac{4}{3}\times\frac{f'(1)}{f'(2)} = \frac{4\times 6}{3\times 2} = 4$

(Q) $f'(2a^+) = \lim_{h\to 0}\frac{f(2a+h) - f(2a)}{h} = 4$ (Given)

Now $f'(-2a^+) = \lim_{h\to 0}\frac{f(-2a+h) - f(-2a)}{h}$

$= \lim_{h\to 0}\frac{f(2a-h) - f(2a)}{h}$ ($\because$ f is even function)

$= \lim_{h\to 0}\frac{f(2a+h) - f(2a)}{h} = f'(2a^+) = 4$

$\because \quad f(x) = f(4a - x)\ \forall\ x \in [2a, 4a]$

Put $x = 2a + h$, we get $f(2a+h) = f(2a-h)$

(R) We have $F(x) = f(e^x)$

$\Rightarrow \quad F'(x) = e^x f'(e^x)$

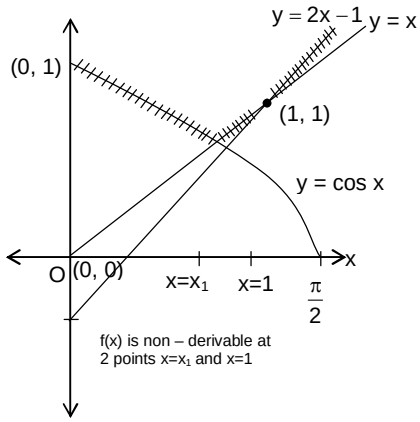
And $G(x) = e^{f(x)}$

$\Rightarrow \quad G'(x) = e^{f(x)}f'(x)$

$\therefore \quad \frac{G'(0)}{F'(0)} = \frac{f'(0)e^{f(0)}}{e^0 f'(e^0)}$

$= \frac{f'(0)e^{f(0)}}{f'(1)} = \frac{3e^3}{e^3} = 3$

(S) $\therefore$ Number of points of non – differentiability =2

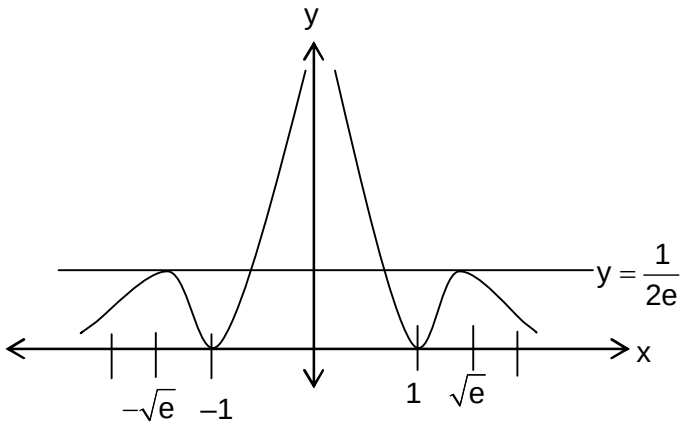


45. C

Sol. $f(x) = |\ln|x|| - kx^2 = 0$

$\Rightarrow k = \dfrac{|\ln|x||}{x^2}$

Consider $g(x) = \dfrac{|\ell n|x||}{x^2}$ as shown



46. D

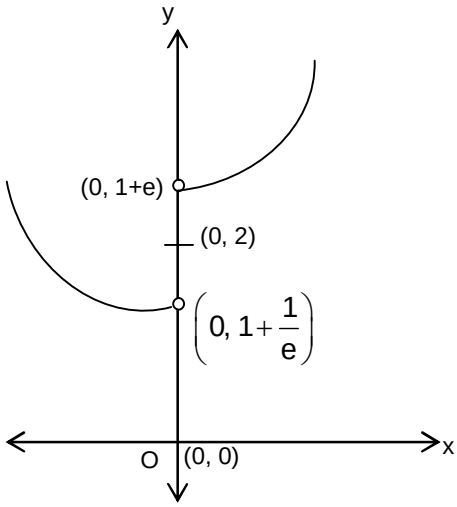
Sol. (P) $f(x) = e^{\text{sgn}\,x} + e^{x^2}$

When $x = 0$ $f(0) = 2$

When $x > 0$ $f(x) = e + e^{x^2}$

When $x < 0$ $f(x) = \dfrac{1}{e} + e^{x^2}$

Hence, $f(x)$ is many – one and neither odd nor even.



(Q) $D_f : 1-x^2 > 0$

$x^2 - 1 < 0 \Rightarrow x \in (-1, 1)$

$\left[x^4\right]$ when $x \in (-1,1)$ is equal to 0.

$\therefore f(x) = \dfrac{1}{\sqrt{1-x^2}}$

Hence, $f(x)$ f is even and many – one.

(R) $f(x) = x^4 + 1 + \left(\dfrac{x^4 + x^2 + 1}{x^2 + x + 1}\right)$

$= x^4 + 1 + x^2 - x + 1$

$= x^4 + x^2 - x + 2 = x\left(x^3 + x - 1\right) + 2$

Hence, $f(x)$ is many – one and neither odd nor even.

(S) $f(x) = x + 3x^3 + 5x^5 + ... + 101x^{101}$

Clearly $f(x)$ is an odd function.

Now, $f'(x) > 0 \forall x \in R$

$\Rightarrow f'(x) > 0 \forall x \in R$

Hence, $f(x)$ f is one – one function

SECTION – D

47. 00000.25

Sol. Put $x\sin y + y\cos y = t$ and after rearrangement put $\sin y = z$

48. 00000.50

Sol. Above sequence is in A.P.

$$\frac{x_1 + 1}{x_1} = \frac{x_2 + 3}{x_2} = ... \frac{x_{1005} + 2009}{x_{1005}} = k$$

$$\therefore k = \frac{(x_1 + 1) + (x_2 + 3) + ...(x_{1005} + 2009)}{x_1 + x_2 .....x_{1005}}$$

$$k = 1 + \frac{(1005)^2}{2010} = \frac{x_{21} + 41}{x_{21}}$$

$$\Rightarrow x_{21} = \frac{82}{1005}$$

49. 00007.56

Sol. Let $\alpha - d, \alpha, \alpha + d$ are roots

$\therefore 3\alpha = -a$ (1)

$3\alpha^2 - d^2 = b$ (2)

$\alpha\left(\alpha^2 - d^2\right) = \dfrac{1}{9}$ (3)

From (1) and (2), $d^2 = \frac{a^2}{3} - b$

Put d^2 in (3) to get

$2a^3 - 9ab + 1 = 4$

50. 00000.22

Sol. Let number of marbles are

Red = a, White = b, Blue = c, Green = d

$\therefore \frac{{}^{a}C_4}{T} = \frac{b\left({}^{a}C_3\right)}{T} = \frac{(b)(c)\left({}^{a}C_4\right)}{T} = \frac{abcd}{T}$

where $T = {}^{a+b+c+d}C_4$

solving, $b = \frac{a-3}{4}$, $c = \frac{a-2}{3}$, $d = \frac{a-1}{2}$

N is max when a = 95

$\therefore N_{max} = 196$

51. 00000.50

Sol. $9A^2 - 6A + I = \begin{bmatrix} 6 & 27 \\ -10 & -1 \end{bmatrix} B^{-1}$

$\Rightarrow (3A - I)^2 = 81$

$\Rightarrow |3A - I| = \pm 9 \quad (1)$

Let $A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$

$\therefore g(ad - bc) = 18$ or 0 (using equation 1)

Since $|A| \neq 0$

$\therefore ad - bc = 2 \Rightarrow |A| = 2$

52. 00000.30

Sol. Let $S = \sum_{k=0}^{100} \left(\frac{k}{k+1}\right) {}^{100}C_k$

$= \sum_{k=0}^{100} \frac{((k+1)-1)}{(k+1)} {}^{100}C_k = \left(\sum_{k=0}^{100} {}^{100}C_k\right) - \sum_{k=0}^{100} \frac{{}^{100}C_k}{k+1}$

$= 2^{100} - \frac{1}{101} \sum_{k=0}^{100} \left(\frac{101}{k+1} {}^{100}C_k\right)$

$= 2^{100} - \frac{1}{101} \sum_{k=0}^{100} {}^{101}C_{k+1}$

$2^{100} - \left(\frac{2^{101} - 1}{101}\right) = \frac{(101)2^{100} - 2^{101} + 1}{101} = \frac{99\left(2^{100}\right) + 1}{101}$

$= \frac{a\left(2^{100}\right) + b}{c}$ (Given)

So, $a = 99, b = 1, c = 101$

Hence, $(a+b+c)_{least} = 99+1+101 = 201.$

53. 00000.14

Sol. Let $I = \lim\limits_{x\to 1}\left(\dfrac{9}{1-x^9} - \dfrac{7}{1-x^7}\right)$ .....(i)

Put $x = \dfrac{1}{y}$, we get $I = \lim\limits_{x\to 1}\left(\dfrac{-9x^9}{1-x^9} + \dfrac{7x^7}{1-x^7}\right)$ .....(ii)

$\therefore$ On adding equation (i) and (ii), we get

$2I = 9 - 7 = 2$

$\Rightarrow I = 1$

Common ratio $= \lim\limits_{x\to 0}\dfrac{\tan x - \sin x}{\ln^3(1+x)}$

$= \lim\limits_{x\to 0}\dfrac{\tan x - \sin x}{x^3\left(\dfrac{\ln(1+x)}{x}\right)^3}$

$= \lim\limits_{x\to 0}\dfrac{\tan x(1-\cos x)}{x^2} = \dfrac{1}{2}$

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Sol. Since f is injective Sp $\dfrac{p}{r} = \dfrac{p-q}{q-r}$

$\Rightarrow \quad pq - pr = rp - rq$

$\Rightarrow \quad 2pr = q(p+r)$ .....(i)

Also, $\dfrac{q}{r} = \dfrac{r}{p} \Rightarrow p, r, q$ are in G.P.

So, let $r = pa, q = pa^2,$ where a is the common ratio of G.P.

Therefore from equation (i), we get

$2.p.pa = pa^2(p+pa)$

$\Rightarrow \quad 2 = a^2 + a$

$\Rightarrow \quad a^2 + a - 2 = 0$

$\Rightarrow (a+2)(a-1) = 0$

$\Rightarrow \quad a = -2, 1$

So, $(p, -2p, 4p)$ and (p, p, p)

(But common ratio = a= 1, is not possible as p, q, r are non – zero distinct quantities.)

Also, $p+q+r = 6$ [As $g(x) = px^2 + qx + r$ passes through M (1, 6)]

$\Rightarrow \quad p + 4p - 2p = 6$

$\Rightarrow \quad p = 2$

Hence $q = 4p = 4(2) = 8$